

20.1 Basic configuration

The I²S interface is configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bit PCI2S.
Remark: On reset, the I²S interface is disabled (PCI2S = 0).
2. Clock: In PCLKSEL1 select PCLK_I2S, see [Table 41](#).
3. Pins: Select I²S pins and their modes in PINSEL0 to PINSEL4 and PINMODE0 to PINMODE4 (see [Section 8.5](#)).
4. Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
5. DMA: The I²S interface supports two DMA requests, see [Table 411](#) and [Table 412](#), and [Table 544](#).

20.2 Features

The I²S-bus provides a standard communication interface for digital audio applications.

The I²S-bus specification defines a 3-wire serial bus, having one data, one clock, and one word select signal. The basic I²S connection has one master, which is always the master, and one slave. The I²S interface on the LPC176x/5x provides a separate transmit and receive channel, each of which can operate as either a master or a slave.

- The I²S input can operate in both master and slave mode.
The I²S output can operate in both master and slave mode, independent of the I²S input.
- Capable of handling 8-bit, 16-bit, and 32-bit word sizes.
- Mono and stereo audio data supported.
- Versatile clocking includes independent transmit and receive fractional rate generators, and an ability to use a single clock input or output for a 4-wire mode.
- The sampling frequency (fs) can range (in practice) from 16 to 96 kHz. (16, 22.05, 32, 44.1, 48, or 96 kHz) for audio applications.
- Separate Master Clock outputs for both transmit and receive channels support a clock up to 512 times the I²S sampling frequency.
- Word Select period in master mode is configurable (separately for I²S input and I²S output).
- Two 8 word (32 byte) FIFO data buffers are provided, one for transmit and one for receive.
- Generates interrupt requests when buffer levels cross a programmable boundary.
- Two DMA requests, controlled by programmable buffer levels. These are connected to the General Purpose DMA block.
- Controls include reset, stop and mute options separately for I²S input and I²S output.

20.3 Description

The I²S performs serial data out via the transmit channel and serial data in via the receive channel. These support the NXP Inter IC Audio format for 8-bit, 16-bit and 32-bit audio data, both for stereo and mono modes. Configuration, data access and control is performed by a APB register set. Data streams are buffered by FIFOs with a depth of 8 words.

The I²S receive and transmit stage can operate independently in either slave or master mode. Within the I²S module the difference between these modes lies in the word select (WS) signal which determines the timing of data transmissions. Data words start on the next falling edge of the transmitting clock after a WS change. In stereo mode when WS is low left data is transmitted and right data when WS is high. In mono mode the same data is transmitted twice, once when WS is low and again when WS is high.

- In master mode, word select is generated internally with a 9-bit counter. The half period count value of this counter can be set in the control register.
- In slave mode, word select is input from the relevant bus pin.
- When an I²S-bus is active, the word select, receive clock and transmit clock signals are sent continuously by the bus master, while data is sent continuously by the transmitter.
- Disabling the I²S can be done with the stop or mute control bits separately for the transmit and receive.
- The stop bit will disable accesses by the transmit channel or the receive channel to the FIFOs and will place the transmit channel in mute mode.
- The mute control bit will place the transmit channel in mute mode. In mute mode, the transmit channel FIFO operates normally, but the output is discarded and replaced by zeroes. This bit does not affect the receive channel, data reception can occur normally.

20.4 Pin descriptions

Table 404. Pin descriptions

Pin Name	Type	Description
I2SRX_CLK	Input/ Output	Receive Clock. A clock signal used to synchronize the transfer of data on the receive channel. It is driven by the master and received by the slave. Corresponds to the signal SCK in the I ² S-bus specification.
I2SRX_WS	Input/ Output	Receive Word Select. Selects the channel from which data is to be received. It is driven by the master and received by the slave. Corresponds to the signal WS in the I ² S-bus specification. WS = 0 indicates that data is being received by channel 1 (left channel). WS = 1 indicates that data is being received by channel 2 (right channel).
I2SRX_SDA	Input/ Output	Receive Data. Serial data, received MSB first. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the I ² S-bus specification.
RX_MCLK	Output	Optional master clock output for the I ² S receive function.
I2STX_CLK	Input/ Output	Transmit Clock. A clock signal used to synchronize the transfer of data on the transmit channel. It is driven by the master and received by the slave. Corresponds to the signal SCK in the I ² S-bus specification.
I2STX_WS	Input/ Output	Transmit Word Select. Selects the channel to which data is being sent. It is driven by the master and received by the slave. Corresponds to the signal WS in the I ² S-bus specification. WS = 0 indicates that data is being sent to channel 1 (left channel). WS = 1 indicates that data is being sent to channel 2 (right channel).
I2STX_SDA	Input/ Output	Transmit Data. Serial data, sent MSB first. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the I ² S-bus specification.
TX_MCLK	Output	Optional master clock output for the I ² S transmit function.

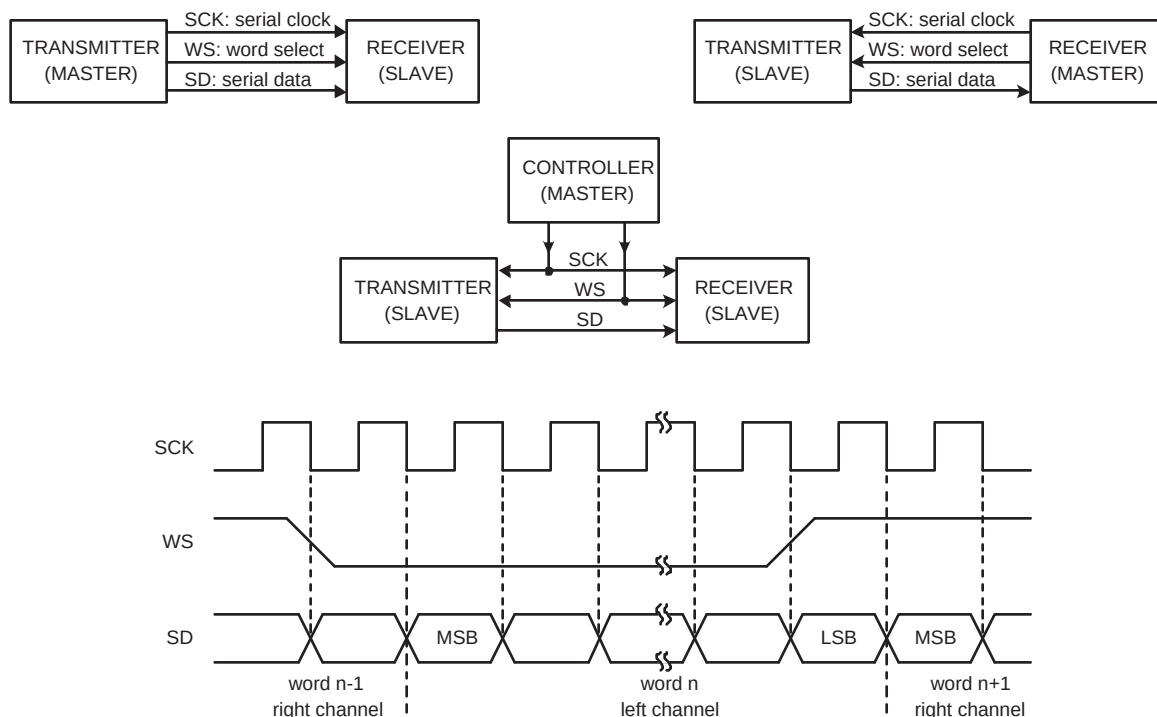


Fig 101. Simple I²S configurations and bus timing

20.5 Register description

[Table 405](#) shows the registers associated with the I²S interface and a summary of their functions. Following the table are details for each register.

Table 405. I²S register map

Name	Description	Access	Reset Value ^[1]	Address
I2SDAO	Digital Audio Output Register. Contains control bits for the I ² S transmit channel.	R/W	0x87E1	0x400A 8000
I2SDAI	Digital Audio Input Register. Contains control bits for the I ² S receive channel.	R/W	0x07E1	0x400A 8004
I2STXFIFO	Transmit FIFO. Access register for the 8 × 32-bit transmitter FIFO.	WO	0	0x400A 8008
I2SRXFIFO	Receive FIFO. Access register for the 8 × 32-bit receiver FIFO.	RO	0	0x400A 800C
I2SSTATE	Status Feedback Register. Contains status information about the I ² S interface.	RO	0x7	0x400A 8010
I2SDMA1	DMA Configuration Register 1. Contains control information for DMA request 1.	R/W	0	0x400A 8014
I2SDMA2	DMA Configuration Register 2. Contains control information for DMA request 2.	R/W	0	0x400A 8018
I2SIRQ	Interrupt Request Control Register. Contains bits that control how the I ² S interrupt request is generated.	R/W	0	0x400A 801C
I2STXRATE	Transmit MCLK divider. This register determines the I ² S TX MCLK rate by specifying the value to divide PCLK by in order to produce MCLK.	R/W	0	0x400A 8020
I2SRXRATE	Receive MCLK divider. This register determines the I ² S RX MCLK rate by specifying the value to divide PCLK by in order to produce MCLK.	R/W	0	0x400A 8024
I2STXBITRATE	Transmit bit rate divider. This register determines the I ² S transmit bit rate by specifying the value to divide TX_MCLK by in order to produce the transmit bit clock.	R/W	0	0x400A 8028
I2SRXBITRATE	Receive bit rate divider. This register determines the I ² S receive bit rate by specifying the value to divide RX_MCLK by in order to produce the receive bit clock.	R/W	0	0x400A 802C
I2STXMODE	Transmit mode control.	R/W	0	0x400A 8030
I2SRXMODE	Receive mode control.	R/W	0	0x400A 8034

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

20.5.1 Digital Audio Output register (I2SDAO - 0x400A 8000)

The I2SDAO register controls the operation of the I²S transmit channel. The function of bits in DAO are shown in [Table 406](#).

Table 406: Digital Audio Output register (I2SDAO - address 0x400A 8000) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	wordwidth		Selects the number of bytes in data as follows:	01
		00	8-bit data	
		01	16-bit data	
		10	Reserved, do not use this setting	
		11	32-bit data	

Table 406: Digital Audio Output register (I2SDAO - address 0x400A 8000) bit description ...continued

Bit	Symbol	Value	Description	Reset Value
2	mono		When 1, data is of monaural format. When 0, the data is in stereo format.	0
3	stop		When 1, disables accesses on FIFOs, places the transmit channel in mute mode.	0
4	reset		When 1, asynchronously resets the transmit channel and FIFO.	0
5	ws_sel		When 0, the interface is in master mode. When 1, the interface is in slave mode. See Section 20.7 for a summary of useful combinations for this bit with I2STXMODE.	1
14:6	ws_halfperiod		Word select half period minus 1, i.e. WS 64clk period -> ws_halfperiod = 31.	0x1F
15	mute		When 1, the transmit channel sends only zeroes.	1
31:16	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.2 Digital Audio Input register (I2SDAI - 0x400A 8004)

The I2SDAI register controls the operation of the I²S receive channel. The function of bits in DAI are shown in [Table 407](#).

Table 407: Digital Audio Input register (I2SDAI - address 0x400A 8004) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	wordwidth		Selects the number of bytes in data as follows:	01
		00	8-bit data	
		01	16-bit data	
		10	Reserved, do not use this setting	
		11	32-bit data	
2	mono		When 1, data is of monaural format. When 0, the data is in stereo format.	0
3	stop		When 1, disables accesses on FIFOs, places the transmit channel in mute mode.	0
4	reset		When 1, asynchronously reset the transmit channel and FIFO.	0
5	ws_sel		When 0, the interface is in master mode. When 1, the interface is in slave mode. See Section 20.7 for a summary of useful combinations for this bit with I2SRXMODE.	1
14:6	ws_halfperiod		Word select half period minus 1, i.e. WS 64clk period -> ws_halfperiod = 31.	0x1F
31:15	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.3 Transmit FIFO register (I2STXFIFO - 0x400A 8008)

The I2STXFIFO register provides access to the transmit FIFO. The function of bits in I2STXFIFO are shown in [Table 408](#).

Table 408: Transmit FIFO register (I2STXFIFO - address 0x400A 8008) bit description

Bit	Symbol	Description	Reset Value
31:0	I2STXFIFO	8 × 32-bit transmit FIFO.	Level = 0

20.5.4 Receive FIFO register (I2SRXFIFO - 0x400A 800C)

The I2SRXFIFO register provides access to the receive FIFO. The function of bits in I2SRXFIFO are shown in [Table 409](#).

Table 409: Receive FIFO register (I2RXFIFO - address 0x400A 800C) bit description

Bit	Symbol	Description	Reset Value
31:0	I2SRXFIFO	8 × 32-bit transmit FIFO.	level = 0

20.5.5 Status Feedback register (I2SSTATE - 0x400A 8010)

The I2SSTATE register provides status information about the I²S interface. The meaning of bits in I2SSTATE are shown in [Table 410](#).

Table 410: Status Feedback register (I2SSTATE - address 0x400A 8010) bit description

Bit	Symbol	Description	Reset Value
0	irq	This bit reflects the presence of Receive Interrupt or Transmit Interrupt. This is determined by comparing the current FIFO levels to the rx_depth_irq and tx_depth_irq fields in the I2SIRQ register.	1
1	dmareq1	This bit reflects the presence of Receive or Transmit DMA Request 1. This is determined by comparing the current FIFO levels to the rx_depth_dma1 and tx_depth_dma1 fields in the I2SDMA1 register.	1
2	dmareq2	This bit reflects the presence of Receive or Transmit DMA Request 2. This is determined by comparing the current FIFO levels to the rx_depth_dma2 and tx_depth_dma2 fields in the I2SDMA2 register.	1
7:3	Unused	Unused.	0
11:8	rx_level	Reflects the current level of the Receive FIFO.	0
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
19:16	tx_level	Reflects the current level of the Transmit FIFO.	0
31:20	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.6 DMA Configuration Register 1 (I2SDMA1 - 0x400A 8014)

The I2SDMA1 register controls the operation of DMA request 1. The function of bits in I2SDMA1 are shown in [Table 411](#). Refer to the General Purpose DMA Controller chapter for details of DMA operation.

Table 411: DMA Configuration register 1 (I2SDMA1 - address 0x400A 8014) bit description

Bit	Symbol	Description	Reset Value
0	rx_dma1_enable	When 1, enables DMA1 for I ² S receive.	0
1	tx_dma1_enable	When 1, enables DMA1 for I ² S transmit.	0
7:2	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0
11:8	rx_depth_dma1	Set the FIFO level that triggers a receive DMA request on DMA1.	0
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
19:16	tx_depth_dma1	Set the FIFO level that triggers a transmit DMA request on DMA1.	0
31:20	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.7 DMA Configuration Register 2 (I2SDMA2 - 0x400A 8018)

The I2SDMA2 register controls the operation of DMA request 2. The function of bits in I2SDMA2 are shown in [Table 406](#).

Table 412: DMA Configuration register 2 (I2SDMA2 - address 0x400A 8018) bit description

Bit	Symbol	Description	Reset Value
0	rx_dma2_enable	When 1, enables DMA1 for I ² S receive.	0
1	tx_dma2_enable	When 1, enables DMA1 for I ² S transmit.	0
7:2	Unused	Unused.	0
11:8	rx_depth_dma2	Set the FIFO level that triggers a receive DMA request on DMA2.	0
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
19:16	tx_depth_dma2	Set the FIFO level that triggers a transmit DMA request on DMA2.	0
31:20	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.8 Interrupt Request Control register (I2SIRQ - 0x400A 801C)

The I2SIRQ register controls the operation of the I²S interrupt request. The function of bits in I2SIRQ are shown in [Table 406](#).

Table 413: Interrupt Request Control register (I2SIRQ - address 0x400A 801C) bit description

Bit	Symbol	Description	Reset Value
0	rx_irq_enable	When 1, enables I ² S receive interrupt.	0
1	tx_irq_enable	When 1, enables I ² S transmit interrupt.	0
7:2	Unused	Unused.	0
11:8	rx_depth_irq	Set the FIFO level on which to create an irq request.	0
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
19:16	tx_depth_irq	Set the FIFO level on which to create an irq request.	0
31:20	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.9 Transmit Clock Rate register (I2STXRATE - 0x400A 8020)

The MCLK rate for the I²S transmitter is determined by the values in the I2STXRATE register. The required I2STXRATE setting depends on the desired audio sample rate desired, the format (stereo/mono) used, and the data size.

The transmitter MCLK rate is generated using a fractional rate generator, dividing down the frequency of PCLK_I2S. Values of the numerator (X) and the denominator (Y) must be chosen to produce a frequency twice that desired for the transmitter MCLK, which must be an integer multiple of the transmitter bit clock rate. Fractional rate generators have some aspects that the user should be aware of when choosing settings. These are discussed in [Section 20.5.9.1](#). The equation for the fractional rate generator is:

$$I2STXMCLK = PCLK_I2S * (X/Y) / 2$$

Note: If the value of X or Y is 0, then no clock is generated. Also, the value of Y must be greater than or equal to X.

Table 414: Transmit Clock Rate register (I2TXRATE - address 0x400A 8020) bit description

Bit	Symbol	Description	Reset Value
7:0	Y_divider	I ² S transmit MCLK rate denominator. This value is used to divide PCLK to produce the transmit MCLK. Eight bits of fractional divide supports a wide range of possibilities. A value of 0 stops the clock.	0
15:8	X_divider	I ² S transmit MCLK rate numerator. This value is used to multiply PCLK by to produce the transmit MCLK. A value of 0 stops the clock. Eight bits of fractional divide supports a wide range of possibilities. Note: the resulting ratio X/Y is divided by 2.	0
31:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.9.1 Notes on fractional rate generators

The nature of a fractional rate generator is that there will be some output jitter with some divide settings. This is because the fractional rate generator is a fully digital function, so output clock transitions are synchronous with the source clock, whereas a theoretical perfect fractional rate may have edges that are not related to the source clock. So, output jitter will not be greater than plus or minus one source clock between consecutive clock edges.

For example, if X = 0x07 and Y = 0x11, the fractional rate generator will output 7 clocks for every 17 (11 hex) input clocks, distributed as evenly as it can. In this example, there is no way to distribute the output clocks in a perfectly even fashion, so some clocks will be longer than others. The output is divided by 2 in order to square it up, which also helps with the jitter. The frequency averages out to exactly $(7/17) / 2$, but some clocks will be a slightly different length than their neighbors. It is possible to avoid jitter entirely by choosing fractions such that X divides evenly into Y, such as 2/4, 2/6, 3/9, 1/N, etc.

20.5.10 Receive Clock Rate register (I2SRXRATE - 0x400A 8024)

The MCLK rate for the I²S receiver is determined by the values in the I2SRXRATE register. The required I2SRXRATE setting depends on the peripheral clock rate (PCLK_I2S) and the desired MCLK rate (such as 256 fs).

The receiver MCLK rate is generated using a fractional rate generator, dividing down the frequency of PCLK_I2S. Values of the numerator (X) and the denominator (Y) must be chosen to produce a frequency twice that desired for the receiver MCLK, which must be an integer multiple of the receiver bit clock rate. Fractional rate generators have some aspects that the user should be aware of when choosing settings. These are discussed in [Section 20.5.9.1](#). The equation for the fractional rate generator is:

$$I2SRXMCLK = PCLK_I2S * (X/Y) / 2$$

Note: If the value of X or Y is 0, then no clock is generated. Also, the value of Y must be greater than or equal to X.

Table 415: Receive Clock Rate register (I2SRXRATE - address 0x400A 8024) bit description

Bit	Symbol	Description	Reset Value
7:0	Y_divider	I ² S receive MCLK rate denominator. This value is used to divide PCLK to produce the receive MCLK. Eight bits of fractional divide supports a wide range of possibilities. A value of 0 stops the clock.	0
15:8	X_divider	I ² S receive MCLK rate numerator. This value is used to multiply PCLK by to produce the receive MCLK. A value of 0 stops the clock. Eight bits of fractional divide supports a wide range of possibilities. Note: the resulting ratio X/Y is divided by 2.	0
31:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.11 Transmit Clock Bit Rate register (I2STXBITRATE - 0x400A 8028)

The bit rate for the I²S transmitter is determined by the value of the I2STXBITRATE register. The value depends on the audio sample rate desired, and the data size and format (stereo/mono) used. For example, a 48 kHz sample rate for 16-bit stereo data requires a bit rate of $48,000 \times 16 \times 2 = 1.536$ MHz.

Table 416: Transmit Clock Rate register (I2TXBITRATE - address 0x400A 8028) bit description

Bit	Symbol	Description	Reset Value
5:0	tx_bitrate	I ² S transmit bit rate. This value plus one is used to divide TX_MCLK to produce the transmit bit clock.	0
31:6	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.12 Receive Clock Bit Rate register (I2SRXBITRATE - 0x400A 802C)

The bit rate for the I²S receiver is determined by the value of the I2SRXBITRATE register. The value depends on the audio sample rate, as well as the data size and format used. The calculation is the same as for I2SRXBITRATE.

Table 417: Receive Clock Rate register (I2SRXBITRATE - address 0x400A 802C) bit description

Bit	Symbol	Description	Reset Value
5:0	rx_bitrate	I ² S receive bit rate. This value plus one is used to divide RX_MCLK to produce the receive bit clock.	0
31:6	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.13 Transmit Mode Control register (I2STXMODE - 0x400A 8030)

The Transmit Mode Control register contains additional controls for transmit clock source, enabling the 4-pin mode, and how MCLK is used. See [Section 20.7](#) for a summary of useful mode combinations.

Table 418: Transmit Mode Control register (I2STXMODE - 0x400A 8030) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	TXCLKSEL		Clock source selection for the transmit bit clock divider.	0
		00	Select the TX fractional rate divider clock output as the source	
		01	Reserved	
		10	Select the RX_MCLK signal as the TX_MCLK clock source	
		11	Reserved	
2	TX4PIN		Transmit 4-pin mode selection. When 1, enables 4-pin mode.	0
3	TXMCENA		Enable for the TX_MCLK output. When 0, output of TX_MCLK is not enabled. When 1, output of TX_MCLK is enabled.	0
31:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.5.14 Receive Mode Control register (I2SRXMODE - 0x400A 8034)

The Receive Mode Control register contains additional controls for receive clock source, enabling the 4-pin mode, and how MCLK is used. See [Section 20.7](#) for a summary of useful mode combinations.

Table 419: Receive Mode Control register (I2SRXMODE - 0x400A 8034) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	RXCLKSEL		Clock source selection for the receive bit clock divider.	0
		00	Select the RX fractional rate divider clock output as the source	
		01	Reserved	
		10	Select the TX_MCLK signal as the RX_MCLK clock source	
		11	Reserved	
2	RX4PIN		Receive 4-pin mode selection. When 1, enables 4-pin mode.	0
3	RXMCENA		Enable for the RX_MCLK output. When 0, output of RX_MCLK is not enabled. When 1, output of RX_MCLK is enabled.	0
31:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

20.6 I²S transmit and receive interfaces

The I²S interface can transmit and receive 8-bit, 16-bit or 32-bit stereo or mono audio information. Some details of I²S implementation are:

- When the FIFO is empty, the transmit channel will repeat transmitting the same data until new data is written to the FIFO.
- When mute is true, the data value 0 is transmitted.
- When mono is false, two successive data words are respectively left and right data.
- Data word length is determined by the wordwidth value in the configuration register. There is a separate wordwidth value for the receive channel and the transmit channel.
 - 0: word is considered to contain four 8-bit data words.
 - 1: word is considered to contain two 16-bit data words.
 - 3: word is considered to contain one 32-bit data word.
- When the transmit FIFO contains insufficient data the transmit channel will repeat transmitting the last data until new data is available. This can occur when the microprocessor or the DMA at some time is unable to provide new data fast enough. Because of this delay in new data there is a need to fill the gap, which is accomplished by continuing to transmit the last sample. The data is not muted as this would produce an noticeable and undesirable effect in the sound.
- The transmit channel and the receive channel only handle 32-bit aligned words, data chunks must be clipped or extended to a multiple of 32 bits.

When switching between data width or modes the I²S must be reset via the reset bit in the control register in order to ensure correct synchronization. It is advisable to set the stop bit also until sufficient data has been written in the transmit FIFO. Note that when stopped data output is muted.

All data accesses to FIFOs are 32 bits. [Figure 114](#) shows the possible data sequences.

A data sample in the FIFO consists of:

- 1×32 bits in 8-bit or 16-bit stereo modes.
- 1×32 bits in mono modes.
- 2×32 bits, first left data, second right data, in 32-bit stereo modes.

Data is read from the transmit FIFO after the falling edge of WS, it will be transferred to the transmit clock domain after the rising edge of WS. On the next falling edge of WS the left data will be loaded in the shift register and transmitted and on the following rising edge of WS the right data is loaded and transmitted.

The receive channel will start receiving data after a change of WS. When word select becomes low it expects this data to be left data, when WS is high received data is expected to be right data. Reception will stop when the bit counter has reached the limit set by wordwidth. On the next change of WS the received data will be stored in the appropriate hold register. When complete data is available it will be written into the receive FIFO.

20.7 I²S operating modes

The clocking and WS usage of the I²S interface is configurable. In addition to master and slave modes, which are independently configurable for the transmitter and the receiver, several different clock sources are possible, including variations that share the clock and/or WS between the transmitter and receiver. This last option allows using I²S with fewer pins, typically four.

Many configurations are possible that are not considered useful, the following tables and figures give details of the configurations that are most likely to be useful.

Table 420: I²S transmit modes

I2SDAO [5]	I2STXMODE [3:0]	Description
0	0 0 0 0	Typical transmitter master mode. See Figure 102 . The I ² S transmit function operates as a master. The transmit clock source is the fractional rate divider. The WS used is the internally generated TX_WS. The TX_MCLK pin is not enabled for output.
0	0 0 1 0	Transmitter master mode sharing the receiver reference clock. See Figure 103 . The I ² S transmit function operates as a master. The transmit clock source is RX_REF. The WS used is the internally generated TX_WS. The TX_MCLK pin is not enabled for output.
0	0 1 0 0	4-wire transmitter master mode sharing the receiver bit clock and WS. See Figure 104 . The I ² S transmit function operates as a master. The transmit clock source is the RX bit clock. The WS used is the internally generated RX_WS. The TX_MCLK pin is not enabled for output.
0	1 0 0 0	Transmitter master mode with TX_MCLK output. See Figure 102 . The I ² S transmit function operates as a master. The transmit clock source is the fractional rate divider. The WS used is the internally generated TX_WS. The TX_MCLK pin is enabled for output.
1	0 0 0 0	Typical transmitter slave mode. See Figure 105 . The I ² S transmit function operates as a slave. The transmit clock source is the TX_CLK pin. The WS used is the TX_WS pin.
1	0 0 1 0	Transmitter slave mode sharing the receiver reference clock. See Figure 106 . The I ² S transmit function operates as a slave. The transmit clock source is RX_REF. The WS used is the TX_WS pin.
1	0 1 0 0	4-wire transmitter slave mode sharing the receiver bit clock and WS. See Figure 107 . The I ² S transmit function operates as a slave. The transmit clock source is the RX bit clock. The WS used is RX_WS ref.

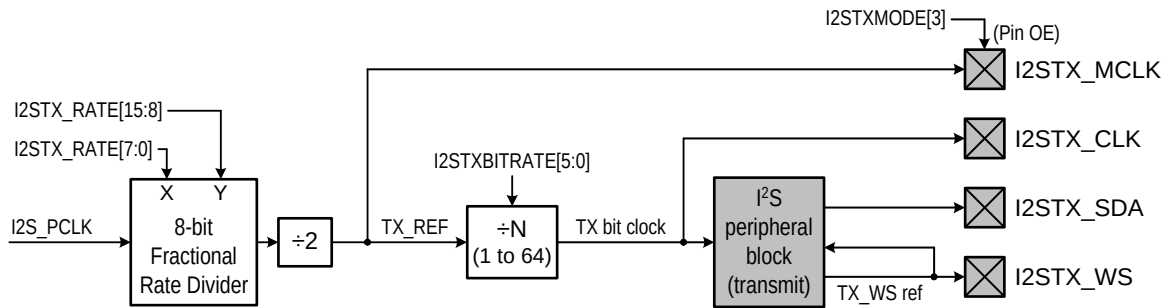


Fig 102. Typical transmitter master mode, with or without MCLK output

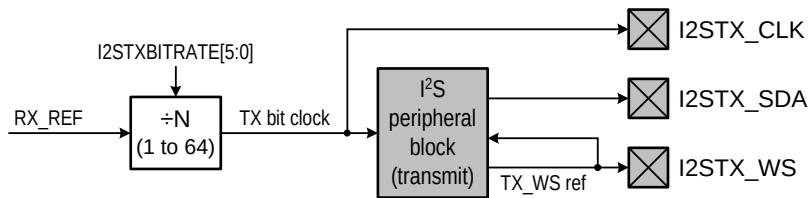


Fig 103. Transmitter master mode sharing the receiver reference clock

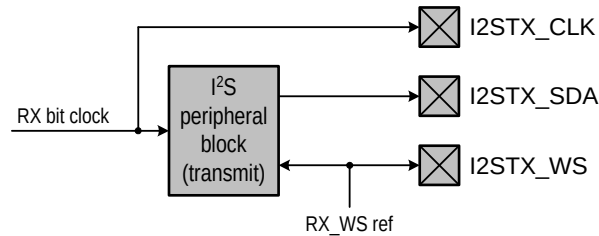


Fig 104. 4-wire transmitter master mode sharing the receiver bit clock and WS

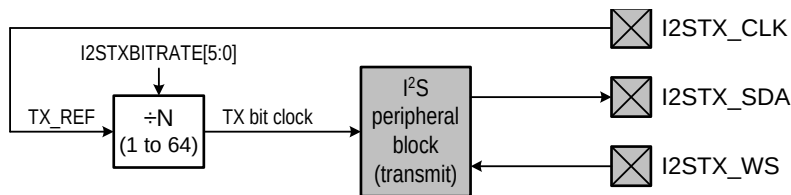


Fig 105. Typical transmitter slave mode

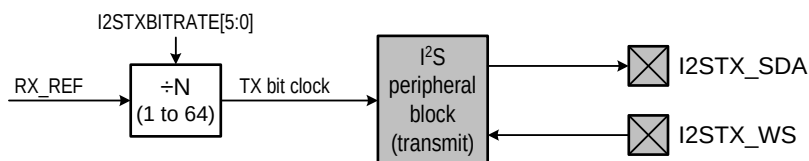


Fig 106. Transmitter slave mode sharing the receiver reference clock

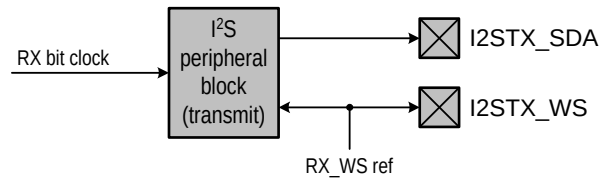


Fig 107. 4-wire transmitter slave mode sharing the receiver bit clock and WS

Table 421: I²S receive modes

I2SDAI [5]	I2SRXMODE [3:0]	Description
0	0 0 0 0	Typical receiver master mode. See Figure 108 . The I ² S receive function operates as a master. The receive clock source is the fractional rate divider. The WS used is the internally generated RX_WS. The RX_MCLK pin is not enabled for output.
0	0 0 1 0	Receiver master mode sharing the transmitter reference clock. See Figure 109 . The I ² S receive function operates as a master. The receive clock source is TX_REF. The WS used is the internally generated RX_WS. The RX_MCLK pin is not enabled for output.
0	0 1 0 0	4-wire receiver master mode sharing the transmitter bit clock and WS. See Figure 110 . The I ² S receive function operates as a master. The receive clock source is the TX bit clock. The WS used is the internally generated TX_WS. The RX_MCLK pin is not enabled for output.
0	1 0 0 0	Receiver master mode with RX_MCLK output. See Figure 108 . The I ² S receive function operates as a master. The receive clock source is the fractional rate divider. The WS used is the internally generated RX_WS. The RX_MCLK pin is enabled for output.
1	0 0 0 0	Typical receiver slave mode. See Figure 111 . The I ² S receive function operates as a slave. The receive clock source is the RX_CLK pin. The WS used is the RX_WS pin.
1	0 0 1 0	Receiver slave mode sharing the transmitter reference clock. See Figure 112 . The I ² S receive function operates as a slave. The receive clock source is TX_REF. The WS used is the RX_WS pin.
1	0 1 0 0	This is a 4-wire receiver slave mode sharing the transmitter bit clock and WS. See Figure 113 . The I ² S receive function operates as a slave. The receive clock source is the TX bit clock. The WS used is TX_WS ref.

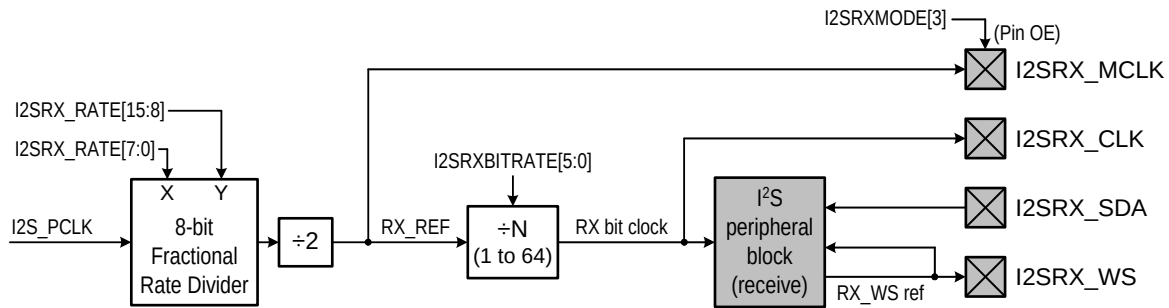


Fig 108. Typical receiver master mode, with or without MCLK output

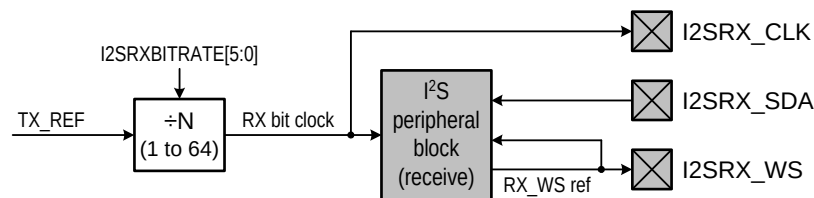


Fig 109. Receiver master mode sharing the transmitter reference clock

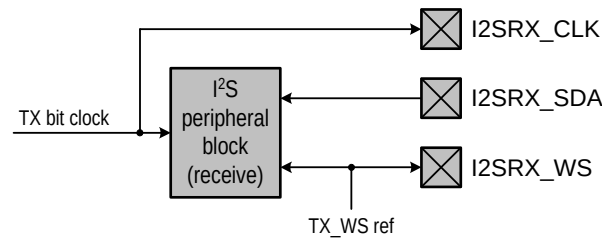


Fig 110. 4-wire receiver master mode sharing the transmitter bit clock and WS

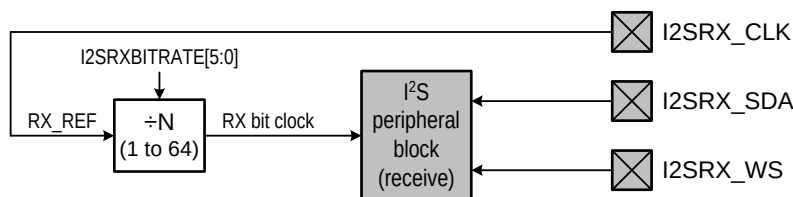


Fig 111. Typical receiver slave mode

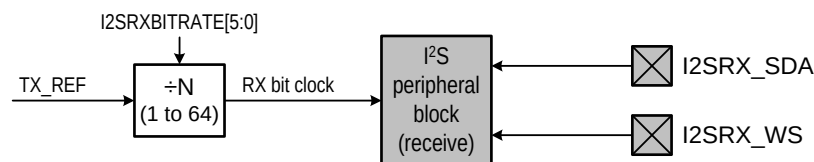


Fig 112. Receiver slave mode sharing the transmitter reference clock

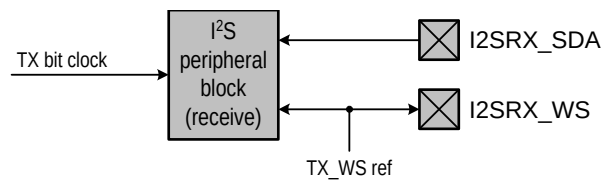


Fig 113. 4-wire receiver slave mode sharing the transmitter bit clock and WS

20.8 FIFO controller

Handling of data for transmission and reception is performed via the FIFO controller which can generate two DMA requests and an interrupt request. The controller consists of a set of comparators which compare FIFO levels with depth settings contained in registers. The current status of the level comparators can be seen in the APB status register.

Table 422. Conditions for FIFO level comparison

Level Comparison	Condition
dmareq_tx_1	tx_depth_dma1 >= tx_level
dmareq_rx_1	rx_depth_dma1 <= rx_level
dmareq_tx_2	tx_depth_dma2 >= tx_level
dmareq_rx_2	rx_depth_dma2 <= rx_level
irq_tx	tx_depth_irq >= tx_level
irq_rx	rx_depth_irq <= rx_level

System signaling occurs when a level detection is true and enabled.

Table 423. DMA and interrupt request generation

System Signaling	Condition
irq	(irq_rx & rx_irq_enable) (irq_tx & tx_irq_enable)
dmareq[0]	(dmareq_tx_1 & tx_dma1_enable) (dmareq_rx_1 & rx_dma1_enable)
dmareq[1]	(dmareq_tx_2 & tx_dma2_enable) (dmareq_rx_2 & rx_dma2_enable)

Table 424. Status feedback in the I2SSTATE register

Status Feedback	Status
irq	irq_rx irq_tx
dmareq1	(dmareq_tx_1 dmareq_rx_1)
dmareq2	(dmareq_rx_2 dmareq_tx_2)

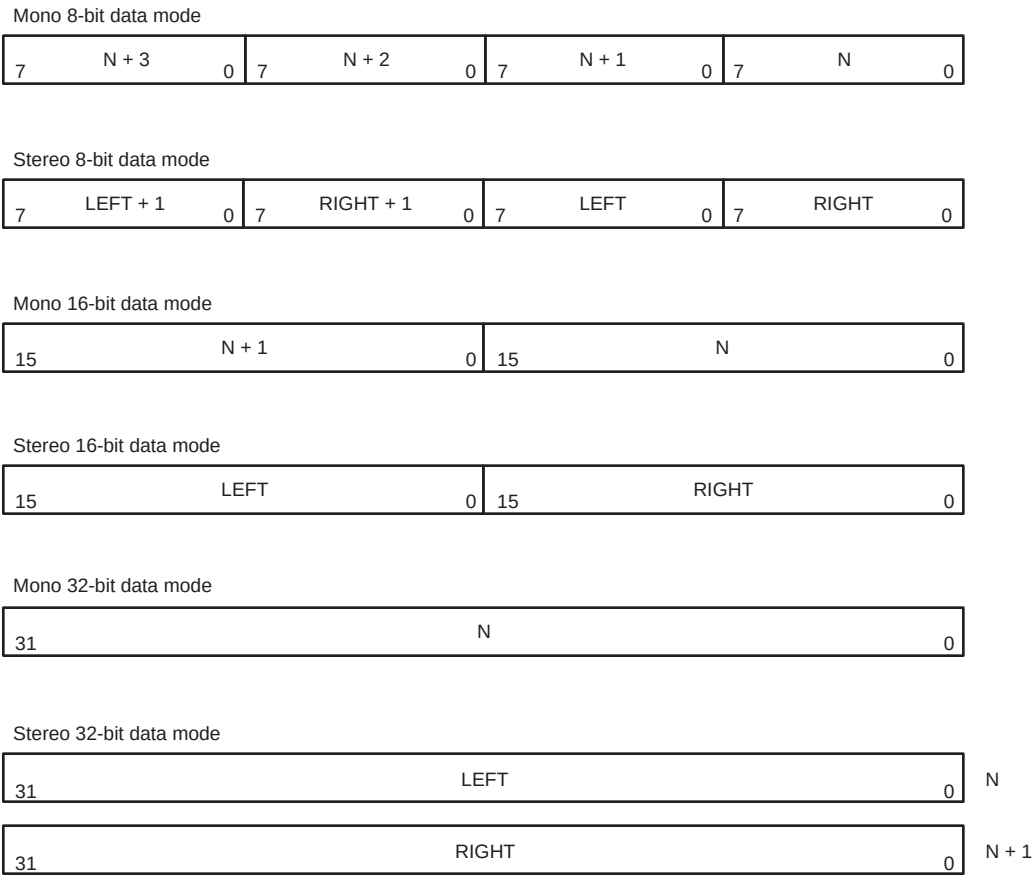


Fig 114. FIFO contents for various I2S modes